

Aortic Dissection and Aneurysmal Disease

Cardiopulmonary-Renal System

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Thank you to former scholars **Andrea Watters, OMS-IV** and **Jesse Chait, DO** whose slides were modified for this lecture

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Make.
Innovate.
Reinvent the Future.

Session Objectives

1. Differentiate the pathogenesis and development of thoracic aortic aneurysms (TAA), aortic abdominal aneurysms (AAA) and aortic dissections.
2. Compare and contrast TAA vs. AAA in their distinct clinical features, risk factors, diagnostic measures, management, and complications
3. Discuss the clinical presentation, diagnosis, classifications and management of aortic dissections.

Lecture Outline

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Anatomy Review

- Aorta and Blood Vessel

What is an Aneurysm?

- Terminology + Classification
- Pathogenesis
- Risk Factors

Thoracic Aortic Aneurysm (TAA)

Abdominal Aortic Aneurysm (AAA)

- Risk factors
- Clinical Presentation & Diagnosis
- Complications
- Management

Aortic Dissection

- Pathogenesis & Classifications
- Clinical Presentations & Diagnosis
- Management

Conclusions

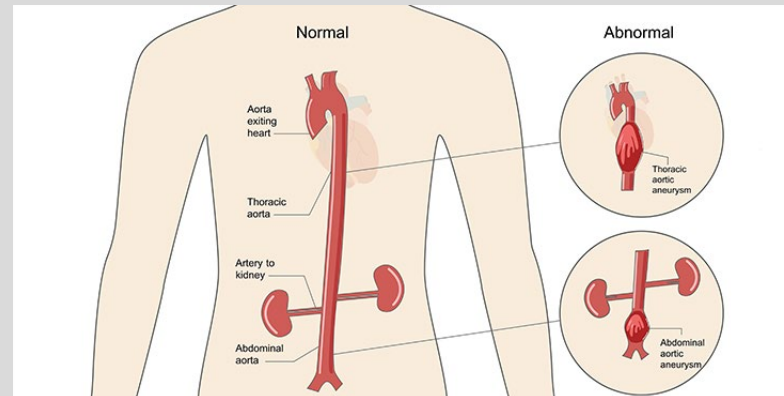
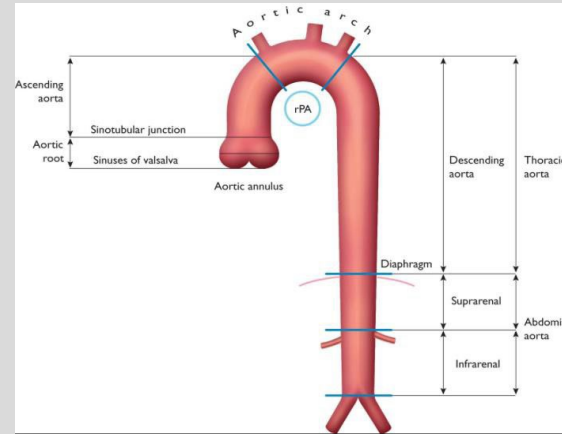
- Summary of lecture content



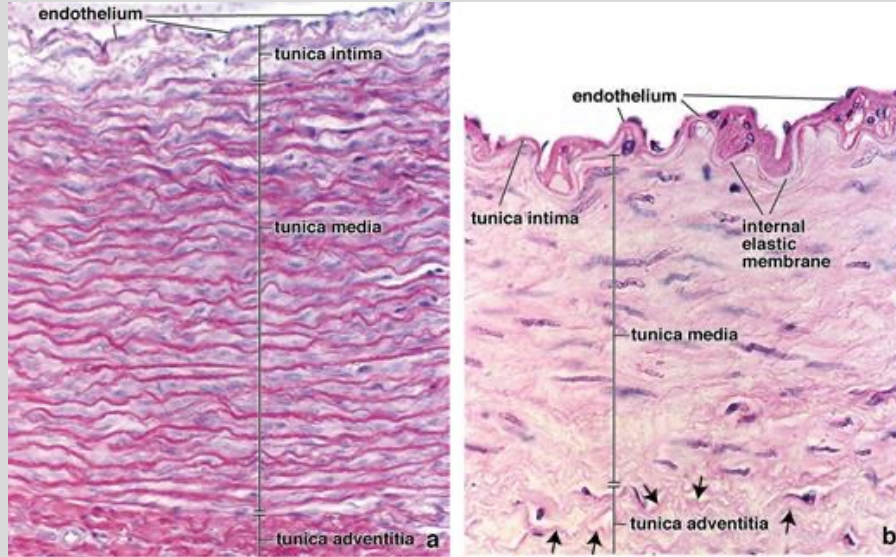
Anatomy Review – Aorta

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- Aortic root
 - Aortic valve + annulus
- Ascending aorta
- Aortic Arch
- Thoracic aorta
- Abdominal
 - Suprarenal
 - Juxtarenal
 - Infrarenal



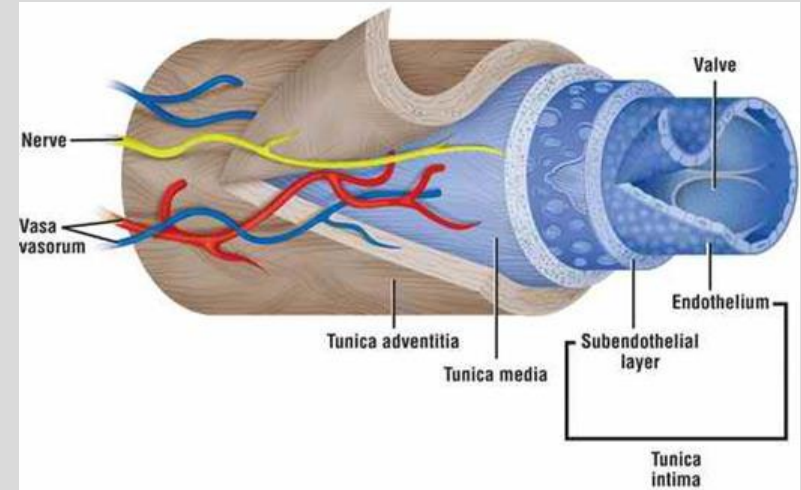
Anatomy Review – Blood Vessels



Think of the aorta as an organ, which requires its own blood supply = **vasa vasorum**

**The infra-renal aorta
lacks a vasa vasorum!**

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- Color Atlas of Text and Histology, 7th edition – Chapter 8
- Histology w/ Correlated Cellular and Molecular Biology, 8th edition – Chapter 13



Normal vs. Abnormal Aorta Size

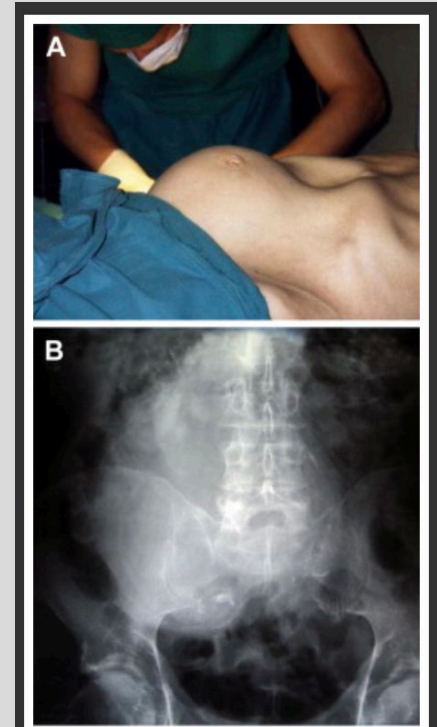
- In most individuals, the diameter of the abdominal aorta is **~2.0cm**
- Largest ever recorded (to my knowledge): **25cm**



Cherry (2cm)

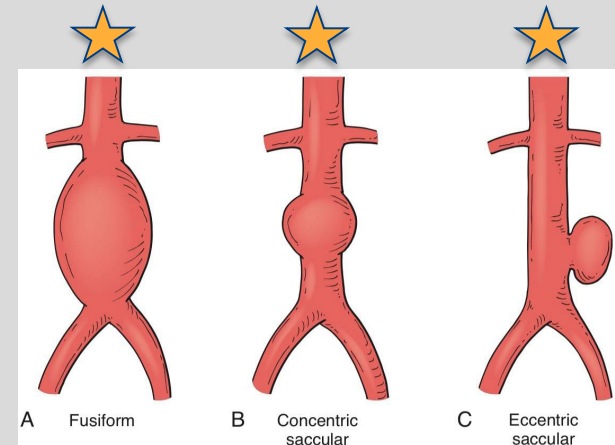


Banana slice (3cm)



Aneurysm Terminology

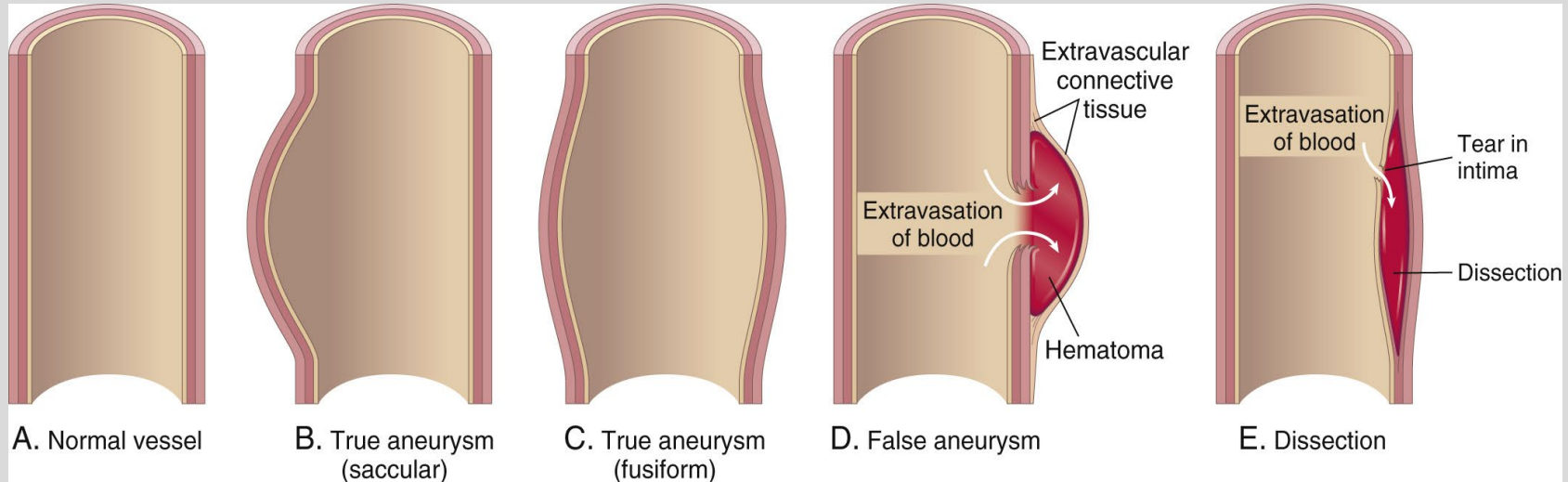
- **Aneurysm:** dilation of a blood vessel >50% normal
 - > 3cm is usually used as the standard; **BUT** depends on what a person's "normal" size is
 - Involves **all three layers** of vessel wall
[Intima/Media/Adventitia]
- **Fusiform aneurysm:** Affects **entire** circumference
- **Saccular aneurysm:** Involves only a **portion** of the circumference
- **Pseudoaneurysm (false aneurysm):** Intima/media layer **disruption** with adventitia intact



Aneurysm Terminology (cont.)

Aneurysm can involve extensive vascular segments:

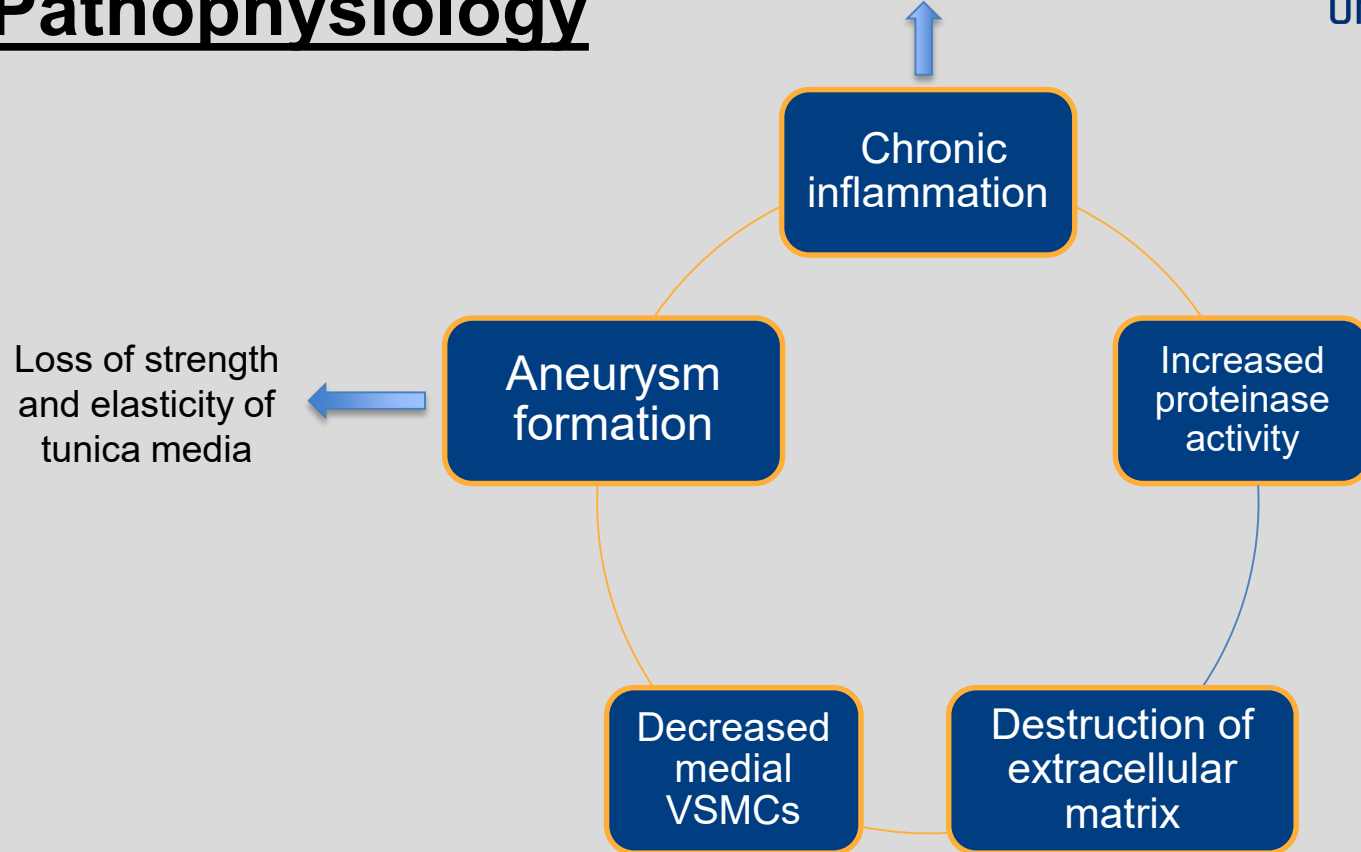
- Thoracic & abdominal aorta, branches, iliac arteries



Aneurysm Pathophysiology

Tobacco (smoking), hypertension
atherosclerosis, genetic predispositions,
systemic inflammatory disease

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Thoracic Aortic Aneurysms (TAA)



TAA – Risk Factors

Degenerative Changes

- **Hypertension**
- Atherosclerosis
- Hypercholesterolemia
- Cigarette smoking
- Infection/Inflammation (syphilis, Staph Aureus)
- Aortitis
- Chest Trauma (ex. car accident)

HyperTension is the most important risk factor for TAA formation!

Congenital Risk Factors (TAA > AAA)

- Marfan Syndrome -> A.D. defect in **fibrillin**
- Ehlers-Danlos Type IV -> A.D. defect in **type-III collagen**
- Loeys-Dietz -> A.D. defect in **TGF-β receptor**
- Bicuspid Aortic Valve



Traumatic Aortic Injury (TAI)

Traumatic rupture of the aorta
due to **acute deceleration**
injury.

“Pseudoaneurysm”

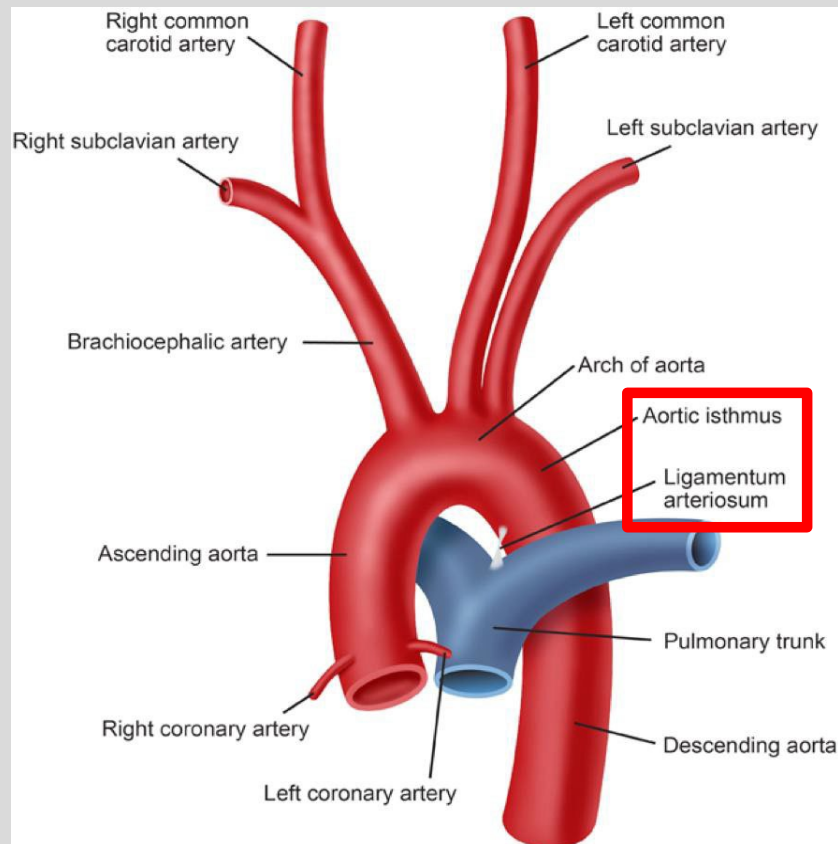
-> High-speed (30+ mph) MVCs or
falls from 10+ feet

**Second leading cause
of traumatic death!**

Most common site is **aortic isthmus** (distal to left subclavian artery).

- Why is this??

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UpToDate: Clinical
Features & Diagnosis
of Blunt Thoracic
Aortic Injury

Marfan Syndrome (MFS)

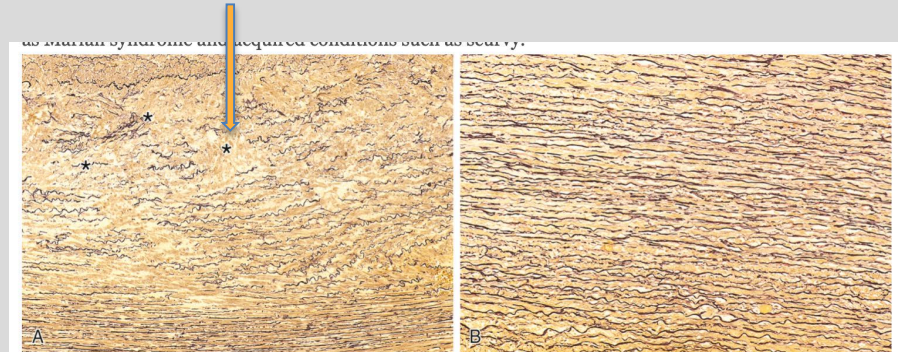
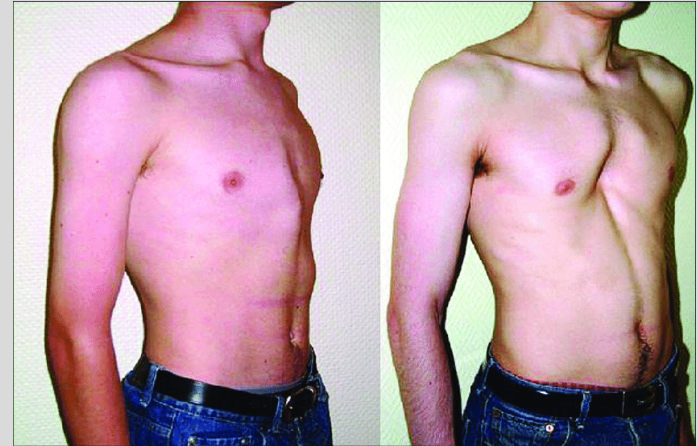
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Pathogenesis of Aneurysms

- **Autosomal dominant** defect in the **fibrillin** gene (chromosome 15)
 - **FBN1 gene mutation**
- **Cystic medial necrosis of aorta**
- Skeleton, eyes, and cardiovascular system = most affected
- **Most common cause of death = aortic root aneurysm rupture or dissection**

Clinical Characteristics

- Wingspan > height
- Long, slender extremities
- **Pectus excavatum or carinatum**
- Upwardly-dislocated lenses
- Mitral valve prolapse
- Scoliosis



Robbins & Cotran Pathologic Basis of Disease, 10th edition

UpToDate: Genetics, clinical features, and diagnosis of Marfan syndrome and related disorders

Ehlers-Danlos Syndrome Type IV (Vascular EDS)

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Pathogenesis of Aneurysms

- Autosomal dominant defect in type-III collagen synthesis
- Type III collagen forms a supportive **mesh** within vessels
- **80% of patients experience a major vascular event or rupture of an internal organ by 40 years old**
 - **Most common cause of death = arterial rupture**
- Life expectancy = 48 years
- At least 1 in 100,000 people (4% of all EDS cases)

Clinical Characteristics

- Thin, translucent skin that bruises easily
- Hypermobility
- Acrogeria (premature aging)
- Large eyes; thin face, lips, and nose
- Poor wound healing



Loeys-Dietz Syndrome (LDS)

Pathogenesis of Aneurysms

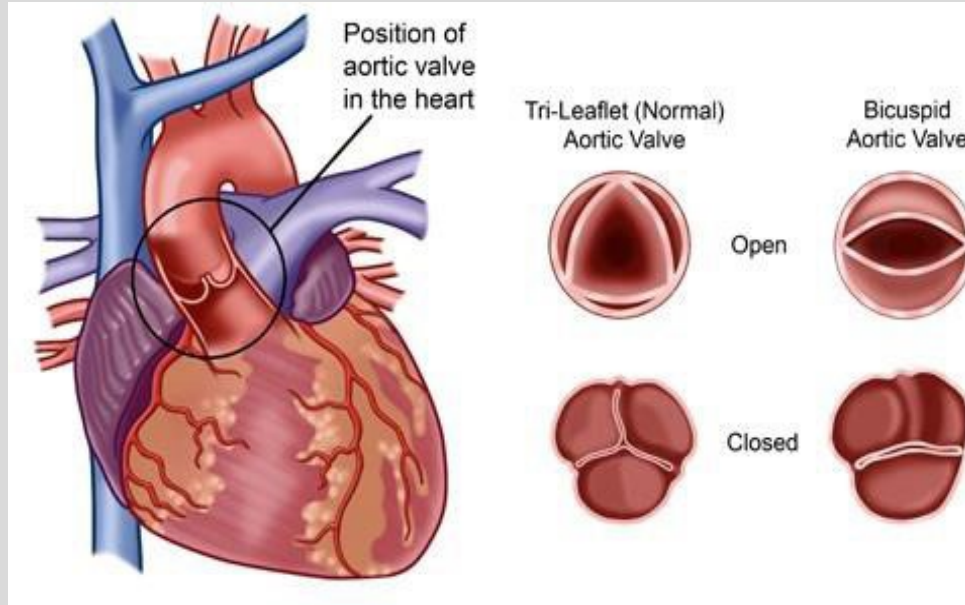
- Autosomal dominant defect in **TGF- β receptor**
- Present with aneurysms much earlier than Marfan & EDS-IV

Clinical Characteristics

- *Often mistaken for Marfan syndrome*
- Features differentiating from Marfan syndrome:
 - Bifid uvula (LDS until proven otherwise)
 - Hypertelorism (90%) = wide-set eyes
 - Cervical spine deformity



Congenital Bicuspid Aortic Valve



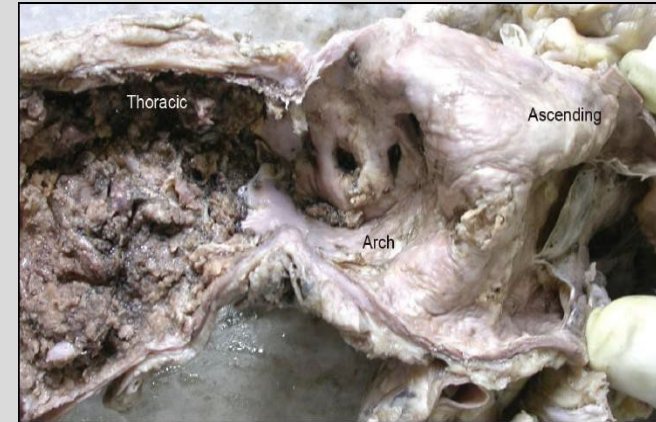
Decreased valve area of bicuspid valve increases aortic pressure → increased wall stress → dilation → ascending TAA formation.

Syphilitic Aortitis

Primary infection of the aortic wall is rare

- Typically 2/2 to another process (endocarditis, IVDA, sepsis)
- Syphilis (*T. pallidum* **endarteritis** of the **vasa vasorum**)
 - Clinical manifestations can occur ~15-30 years after initial infection
 - Results in **obliterative endarteritis** -> interruption of blood flow -> medial ischemia -> loss of elastic & muscular elements -> ANEURYSM

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Syphilitic aortitis -> 3° syphilis and can lead TAA in a patient



TAA – Clinical Presentation

TAA's are usually **silent** unless a complication occurs, or an imaging study has been performed for another reason

- Complications = **rupture** or dissection

The most common presenting symptom of TAA is pain in chest or abdomen.

TAA's can affect surrounding mediastinal structures w/ resulting symptoms:

- **Ascending Thoracic Aorta**
 - Aortic regurgitation due to dilation of aortic root & annulus (can lead to heart failure)
- **Aortic Arch**
 - Coronary artery -> myocardial ischemia or infarction
 - Superior vena cava -> JVD, upper extremity & face edema
 - Pulmonary artery -> pulmonary embolism-like symptoms
 - Airway -> dyspnea, cough, wheezing
 - Left recurrent laryngeal nerve -> hoarseness
- **Descending Thoracic Aorta**
 - Aneurysm eroding into spine -> Back pain

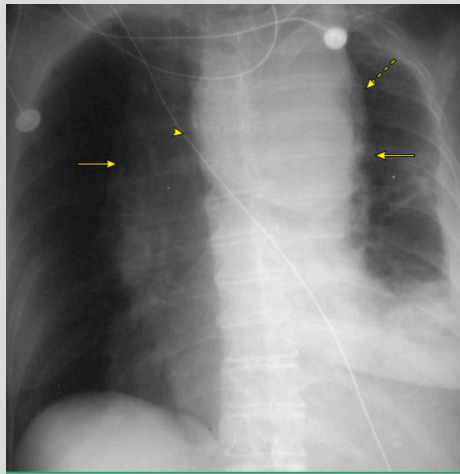


TAA – Diagnosis

Many are discovered incidentally on imaging performed for other reasons!

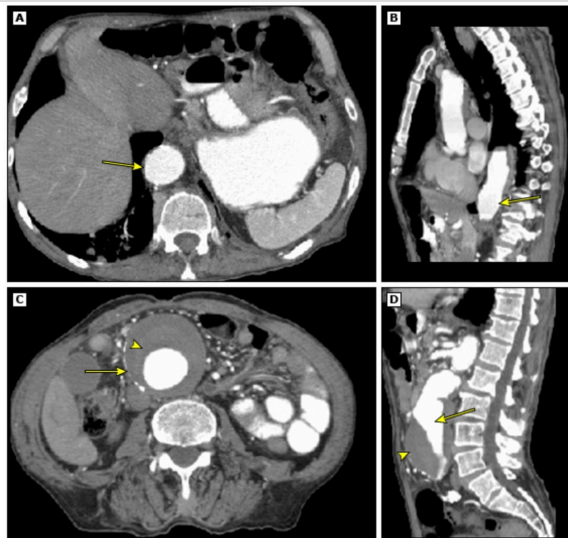
Best initial test (if stable) → CXR

- Widened mediastinum
- Enlargement of aortic knob
- Deviation of trachea



Gold standard → CT-angiogram (CTA)

- Can visualize entire aorta
- **3-D reconstruction/mapping**
- Operative planning
- Can be performed quickly in ED





TAA – Management w/o Repair

Imaging

- CT angiography
- **Every 6 or 12 months**
- Assessing for other aneurysms
- Elective repair not recommended if TAA below certain diameter

20% of patients diagnosed with TAA have AAA

Genetic testing -> TAA can be associated with genetic diseases

If patient is asymptomatic (and diameter <5.5cm):

- Reduce cardiovascular risk
- Manage hypertension (BB > ACE or ARB)



TAA – Management w/ Repair

Repair and Surgical Intervention is indicated if:

- Asymptomatic patient with diameter > **5.5cm**
- Rapid expansion of TAA (>0.5cm/year)
- MFS, LDS, BAV or other inherited disorders
- Patients undergoing aortic valve surgery

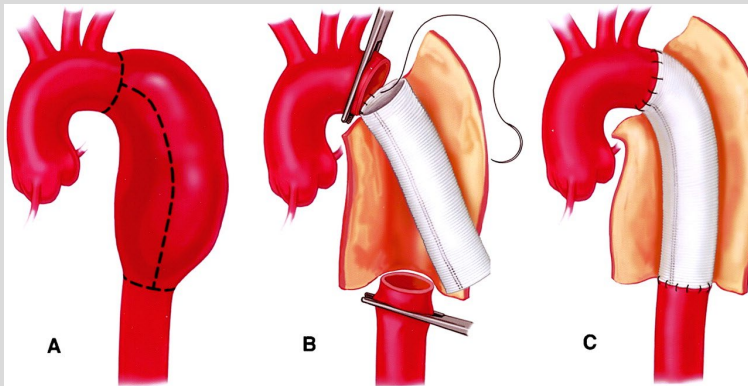
If patient is symptomatic:

- **Surgical intervention (non-ruptured or not)**
- Open or endovascular surgical approach
 - **Open** often requires cardiac bypass
 - **Endovascular** associated with lower perioperative mortality

Marfan syndrome (MFS), familial thoracic aortic aneurysm/dissection (FTAAD), others*	
Without risk factors ^{¶Δ}	≥5.0 cm
With risk factors ^{¶Δ}	≥4.5 cm
Loeys-Dietz syndrome (LDS)	
Without risk factors ^{¶Δ}	≥4.5 cm
With risk factors ^{¶Δ}	≥4.0 cm
Bicuspid aortic valve (BAV)	
Without risk factors ^{¶Δ}	≥5.5 cm
With risk factors ^{¶Δ}	≥5.0 cm
Concomitant aortic valve procedure	≥4.5 cm

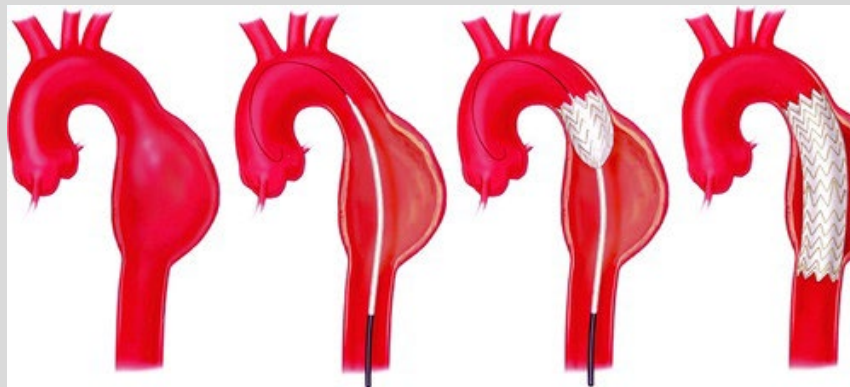
TAA – Surgical repair

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Open

Endovascular



Complications

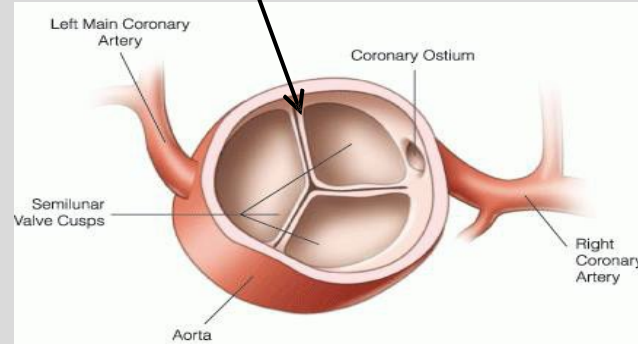
1) Aortic Regurgitation

- Ascending TAAs can **displace the aortic valve commissures**
- Induces annular dilation and **worsens** aortic valve regurgitation

2) Distal Embolization

- TAAs, especially those of the **descending aorta**, are susceptible to thrombus formation
- Those thrombi can **embolize** and travel downstream and occlude:
 - Visceral arteries (i.e. SMA, renal arteries)
 - Lower extremity arteries -> “Blue toe syndrome”

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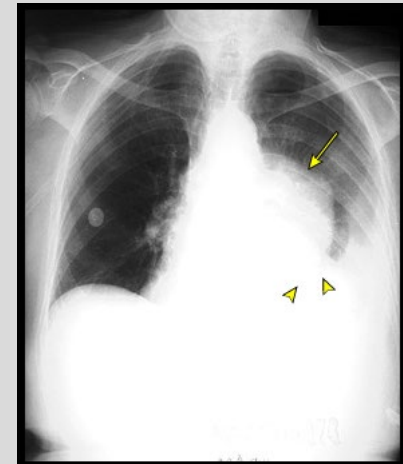
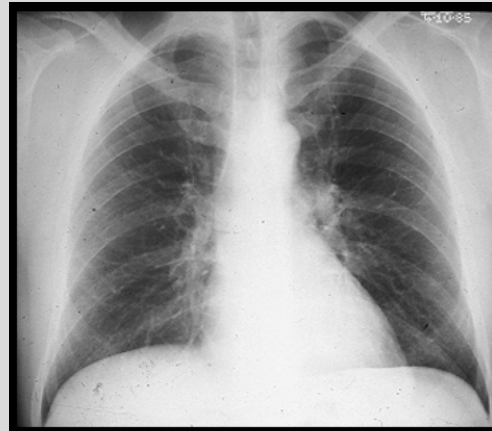
Complications – Rupture

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Patients are **hypotensive** & experience **severe pain** based on location:

- **Anterior chest** → ascending aorta
 - Rupture causes back-bleed into pericardial sack → **acute pericardial tamponade**
- **Upper back** → thoracic aorta
 - Blood pours into pleural cavity
 - Pleural effusion on CXR
 - Hemorrhagic shock
 - Respiratory failure
- **Aortoesophageal fistula** -> esophagus
 - Hematemesis

Mortality is very high in patients that experience a ruptured TAA





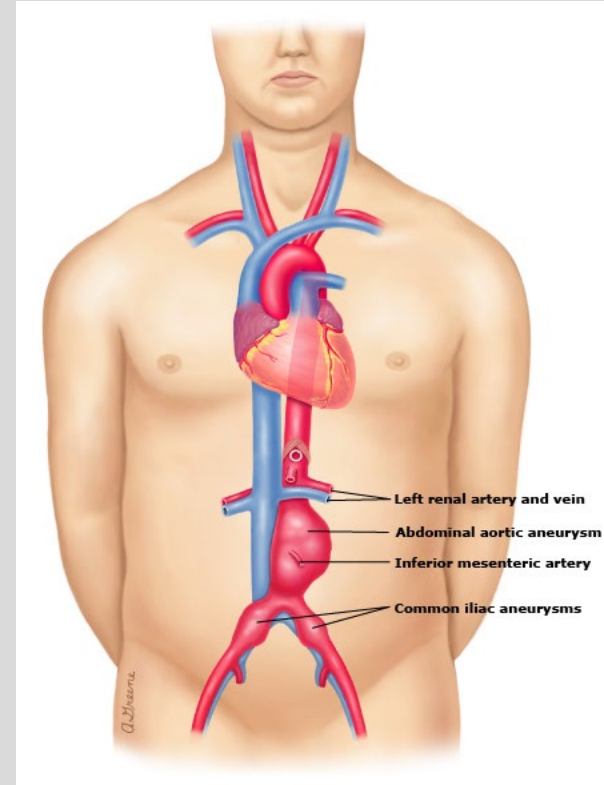
Question Break #1

A 60-year-old man presents to your office for his annual physical. He states that overall he feels fine, he just sometimes gets heart palpitations. He currently takes 20mg of Atorvastatin. Vital signs reveal BMI 30, BP 150/95, pulse 90bpm, temp. 96.8 F, and RR 18 bpm. Office EKG was within normal limits so you order an ultrasound, and incidentally find a thoracic aortic aneurysm (TAA) with a diameter measuring 6.5 cm.

Which of the following would be the next best step?

- A) Refer to vascular surgery for pre-operative assessment
- B) Increase patient's statin and start on aspirin
- C) Recommend lifestyle changes such as walking program
- D) Follow-up within 6 months
- E) Reassurance

Abdominal Aortic Aneurysms (AAA)



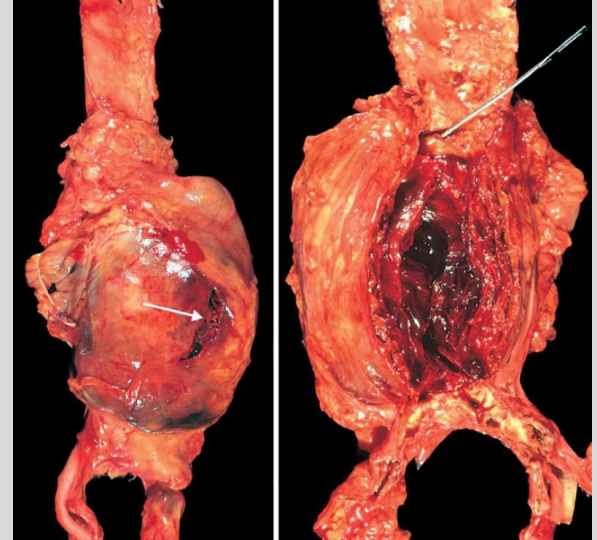
AAA – Pathogenesis

Atherosclerosis is the primary etiology of AAA

- Intimal plaque compresses underlying media -> cuts off oxygen and nutrient diffusion
- Plaques lead to **chronic inflammation** → MMP proliferation → elastin & collagen digested → **wall weakness**
- Thick plaque → no O₂ diffusion → **medial necrosis** → **wall weakness**

Why is the infrarenal aorta most common site of AAA?

- Lacks vasa vasorum and therefore NEEDS diffusion through intima for oxygen + nutrients



Reminder: The abdominal aorta is the vessel most affected by atherosclerosis in the entire body!



AAA – Risk Factors

Degenerative Changes

- **Atherosclerosis**
- **Tobacco use**
- Hypertension
- Hyperlipidemia

Unmodifiable risk factors

- Age (65-80)
- Male sex
- Family history

Smoking is the most important modifiable risk factor for AAA formation!



AAA – Other Variants

Inflammatory AAAs are characterized by dense, periaortic (esp. adventitia) fibrosis and chronic inflammation with **macrophages** and **giant cells**.

- **Can cause IAAA to adhere to adjacent structures (ex. duodenum, ureters)**

Mycotic AAAs are formed due to infection.
Suppurative inflammation destroys the media, furthering dilation → rupture

**Despite the name,
most *mycotic* AAAs
are due to bacteria!**

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Mycotic AAA Microbiology

- *S. aureus* → most common
- *Salmonella* → 2/2 gastroenteritis
- *S. epidermidis* → existing AAAs

AAA – Clinical Features

Like TAAs, most patients with AAA remain **asymptomatic**.

- Physical exam *may* reveal a **pulsatile abdominal mass**.
- Many AAAs are discovered incidentally

Symptomatic AAAs may present as abdominal, flank or back pain

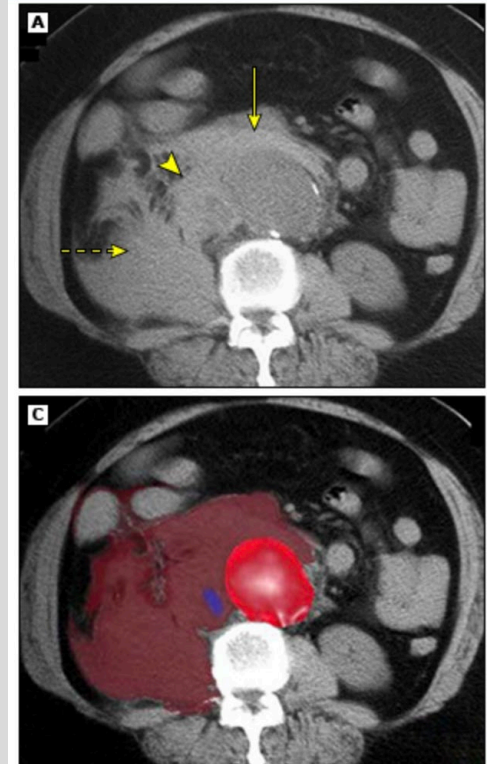
- Complications: May compress local structures, distal embolization

Ruptured AAAs present with the triad of:

- Hypotension
- Flank pain → retroperitoneal irritation due to blood spillage
- Pulsatile abdominal mass

Ruptured AAAs can be contained by retroperitoneum → prolonged survival → chance of repair if caught early

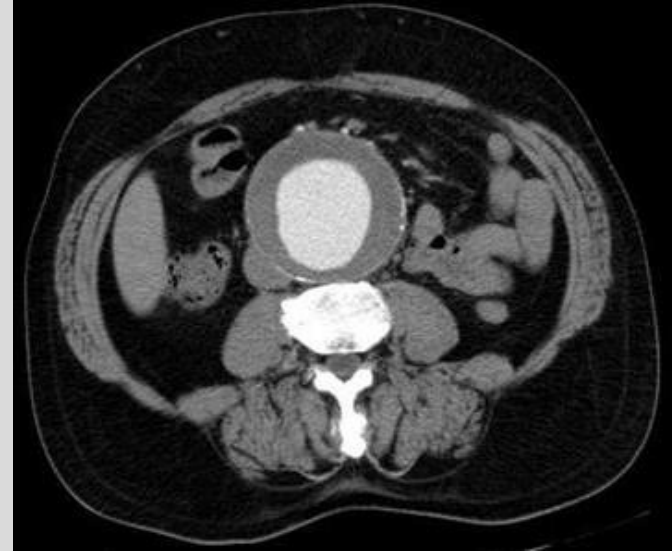
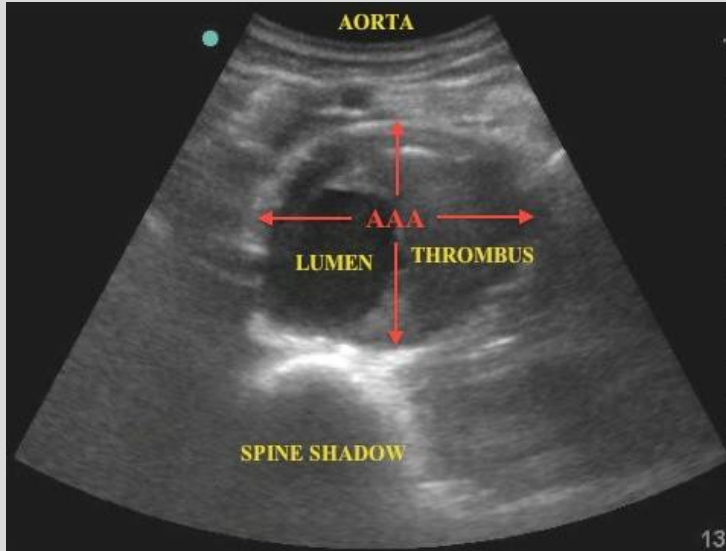
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AAA – Diagnosis

- **Ultrasound** is used to **screen, diagnose,** and **monitor** AAAs
- **CT angiogram** is used to **map** and **plan for intervention.**





AAA Management

- Indication for repair is determined based on **size** & presence of **symptoms**
 - >5.5cm in men; >5cm in women
 - Rapid expansion ≥ 0.5 cm over 6 months or ≥ 1 cm per year
- **Non-ruptured, asymptomatic patient**
 - Manage **hypertension** and **atherosclerosis**
 - Monitor AAA size and growth with regular ultrasounds
- **Non-ruptured, symptomatic patient**
 - Assessing risk for rupture and if patient is a candidate
- **Ruptured, symptomatic patient**
 - Abdominal CT aortography before surgery if patient is hemodynamically stable
 - Endovascular vs. open surgical repair



Question Break #2

A 65-year-old male comes to the clinic for his annual physical. He has a history of hyperlipidemia, hypertension, and diabetes mellitus type 2, all controlled with medication. He states that he drinks 2 "tall boy" beers per day and smokes half a pack a day, with a 30-pack year history. He undergoes an abdominal ultrasound for screening per USPSTF guidelines, which finds a 3.0cm infrarenal abdominal aortic aneurysm (AAA).

Which of the following is most associated with AAA?

- A. Hyperlipidemia
- B. Hypertension
- C. Diabetes mellitus
- D. Cigarette smoking
- E. Excessive alcohol use

Aortic Dissection

Aortic Dissection – Pathogenesis

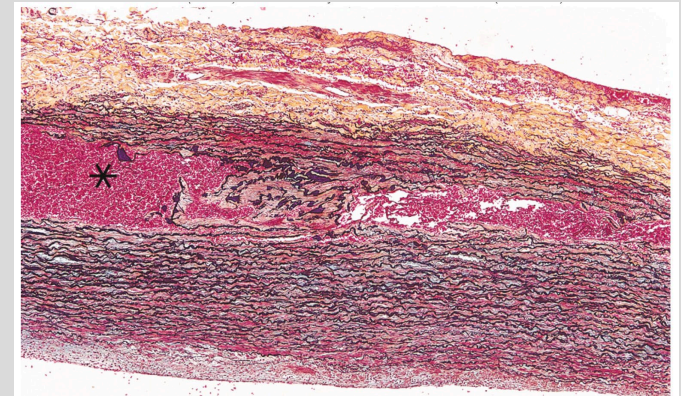
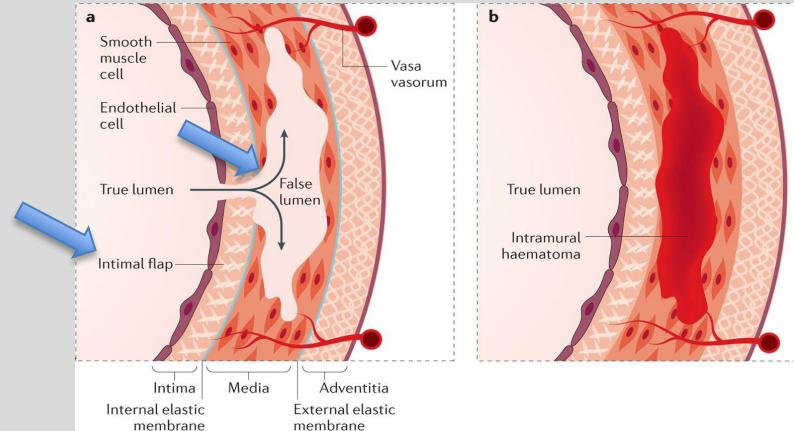
Tearing of intima and inner media from rest of aortic wall, with blood filling in the resulting space.

Why does hypertension contribute?

- Hypertension -> **degenerative changes** with loss of smooth muscle cells and altered ECM content -> **weaker** aorta walls -> tear

Atherosclerosis is NOT a part of the dissection process

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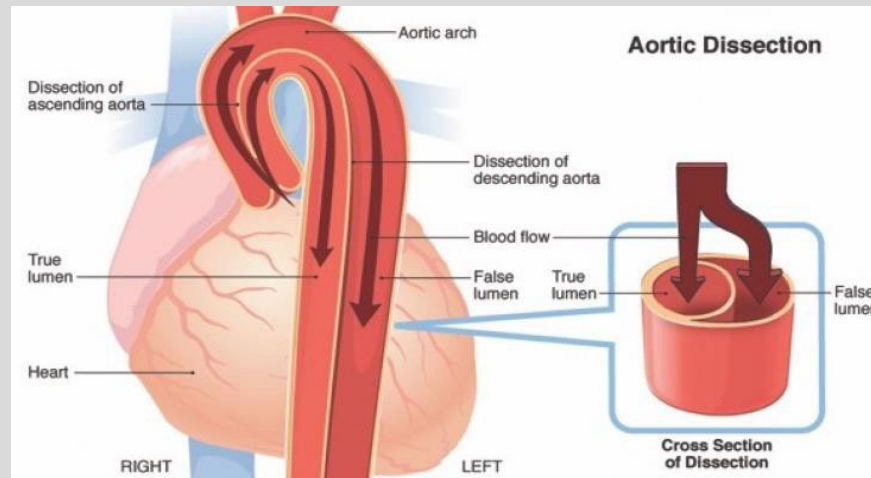


Aortic Dissection – Pathogenesis

Chronic hypertension is the major risk factor for dissection.

Tearing of intima and inner media from rest of aortic wall.

- New cavity is formed → **false lumen**, with can propagate distally **AND** proximally





Aortic Dissection – Risk Factors

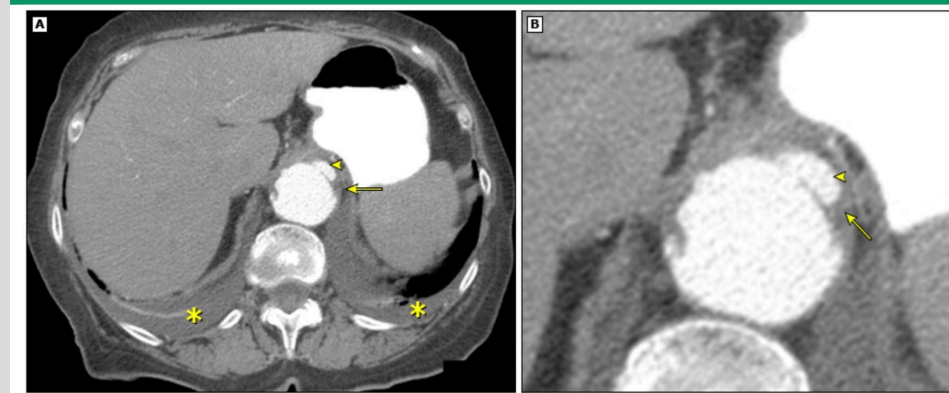
- (1) Men 40-60 years old with chronic hypertension
- (2) Younger patients with connective tissue disorders
 - *Dissection = leading cause of death in Marfan Syndrome*

Additionally:

- Dissections can be iatrogenic; ex. During cardiac catheterization, cardiopulmonary bypass

Aortic Dissection – Clinical Presentation & Diagnosis

- Sudden onset of **severe chest pain**, described as **“tearing”** with **radiation to the back**
 - Pain often severe enough to be confused as myocardial infarction
 - Dissection can propagate into coronaries -> MI
- **Discrepancy in BP** (>20 mmHg) or **pulse in extremities** -> propagation of dissection into subclavian artery



Initial **CXR** will show **widened mediastinum**.

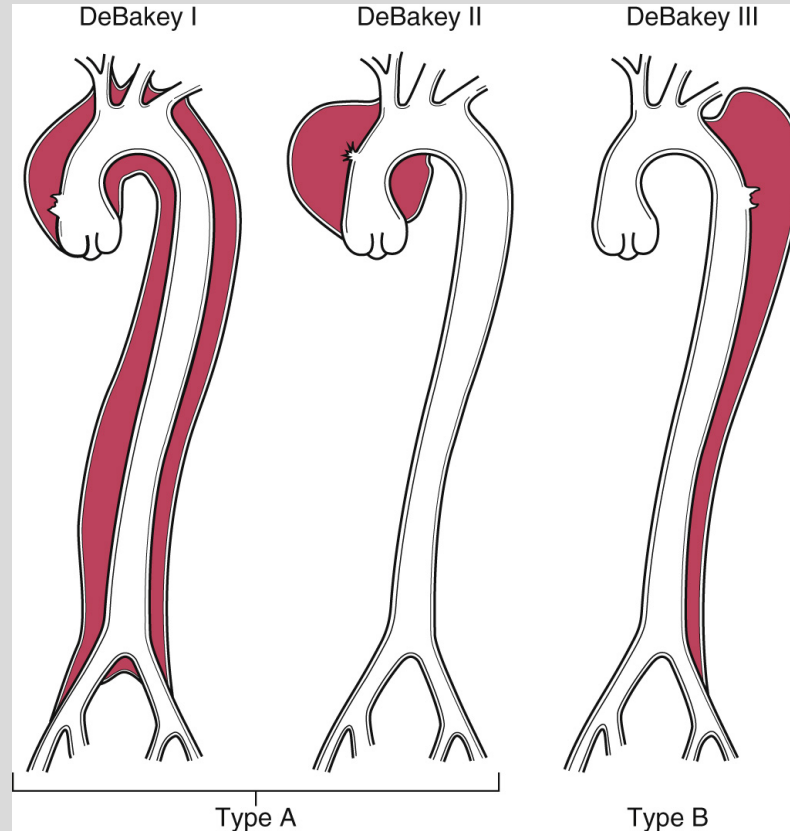
Diagnosis is confirmed with **CT-angiogram (CTA)** if possible.

Trans-esophageal echo (TEE) can be performed emergently.

Aortic Dissection – Classification

Systems

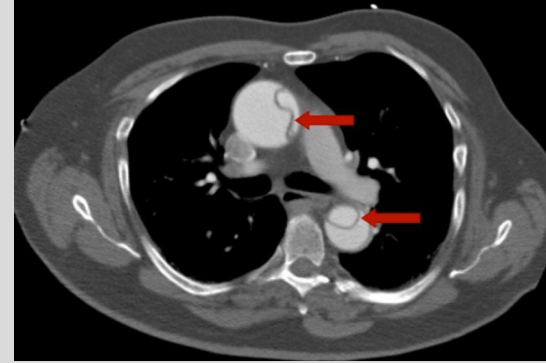
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Aortic Dissection – Classification & Treatment

Stanford Type A involves the
Ascending aorta

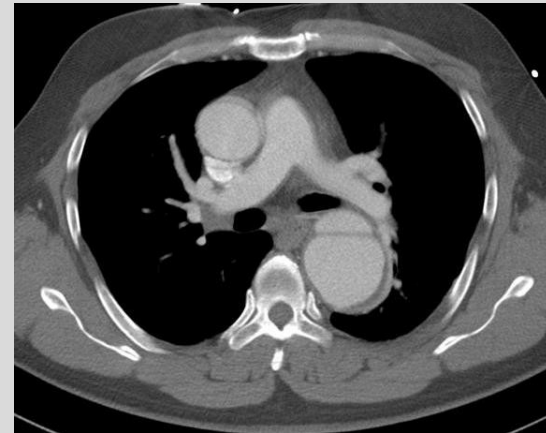
- Can involve the entire aorta
- Pericardium & coronaries involved
- Treat with **emergent surgery**



Type A Dissection

Stanford Type B does **NOT** involve the ascending aorta

- Distal to the left subclavian artery
- Treat initially with **Beta blockers**
 - Goal is to lower BP and prevent propagation
 - Vasodi**L**ators **L**ater
- Surgery is indicated with worsening vitals or complications



Type B Dissection



Aortic Dissection – Complications

Death from aortic dissection is most often related to propagation and rupture of the dissection into the:

- **Pericardium** -> acute pericardial tamponade
- **Aortic valvular annulus** -> aortic insufficiency

Pericardial tamponade and aortic regurgitation are the most common causes of death in a *Type A dissection*!

Dissection of intimal tear can propagate into any aortic branch vessel and cause malperfusion syndrome:

- **Coronary arteries** → myocardial ischemia & infarction
- **Carotid arteries** → TIA, stroke, dysphasia, neurologic deficit
- **Mesenteric arteries** → abdominal pain, mesenteric ischemia
- **Renal arteries** → renal failure, oliguria, anuria



Question Break #3

A 23-year-old male presents to the ED with severe “tearing” chest pain that radiates to the back. Vitals are noted to be 170/100mm Hg in right arm and 90/60 in his left arm. Physical examination notes a tall young man with long arms and the following finding on chest examination.

Which of the following diagnoses most likely predisposed this patient to his current condition?

- A. Cocaine addiction
- B. Marfan Syndrome
- C. Loeys-Dietz Syndrome
- D. Ehler’s Danlos Syndrome
- E. Syphilis



Final Overview

Aneurysms – General Characteristics & Pathophysiology

- Dilation of vessel wall
 - True = all three layers
 - Fusiform – circumferential/concentric; saccular - eccentric
 - False = adventitia intact
- Related to atherosclerosis -> chronic inflammation -> wall weakness
- Matrix metalloproteinases (MMPs) degrade collagen and elastin
- Cystic medial degradation is the primary pathophysiology

Thoracic Aortic Aneurysm (TAA)

- **Risk Factors:** Hypertension, genetically-mediated or congenital
- Syphilitic aortitis (3° syphilis) -> TAA 2/2 vasa vasorum endarteritis
- Usually asymptomatic, may compress mediastinal structures
- Aortic regurgitation due to dilation of aortic root + valve annulus
- Distal embolization may occlude downstream vessels
- Rupture is almost always fatal
- Diagnosis with CXR and/or CTA; typically found incidentally

Final Overview (cont.)

Abdominal Aortic Aneurysm (AAA)

- **Risk factors:** Atherosclerosis, smoking, older age, male sex, family history
- Inflammatory and mycotic AAA= macrophages & *S. aureus*, respectively
- AAAs are typically asymptomatic until rupture or discovered incidentally
- Ruptured AAA (rAAA) = hypotension, flank pain, pulsatile abdominal mass
- Ultrasound used to screen, CTA for mapping & planning of intervention

Aortic Dissection

- Chronic hypertension (in older men) is the leading cause
- Can occur in younger patients with connective tissue disease (i.e. Marfan)
- Tearing of intima + inner media = formation of intimal flap + false lumen
- Presents with chest pain that radiates to the back + pulse/BP deficit + widened mediastinum on CXR
- Stanford Type A = involves ascending → emergent surgery
- Stanford Type B = no ascending involvement → lower BP (BBs) & monitor
- Diagnose with CXR, CTA, or TEE (if patient is too unstable for CT scan)
- Can rupture → tamponade; Propagates → AR, MI, CVA, renal failure, etc.

Thank you!

Email: kkeever@nyit.edu

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Lecture Feedback Form:

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Do.
Make.
Innovate.
Reinvent the Future.